# DP-600

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## Introduction

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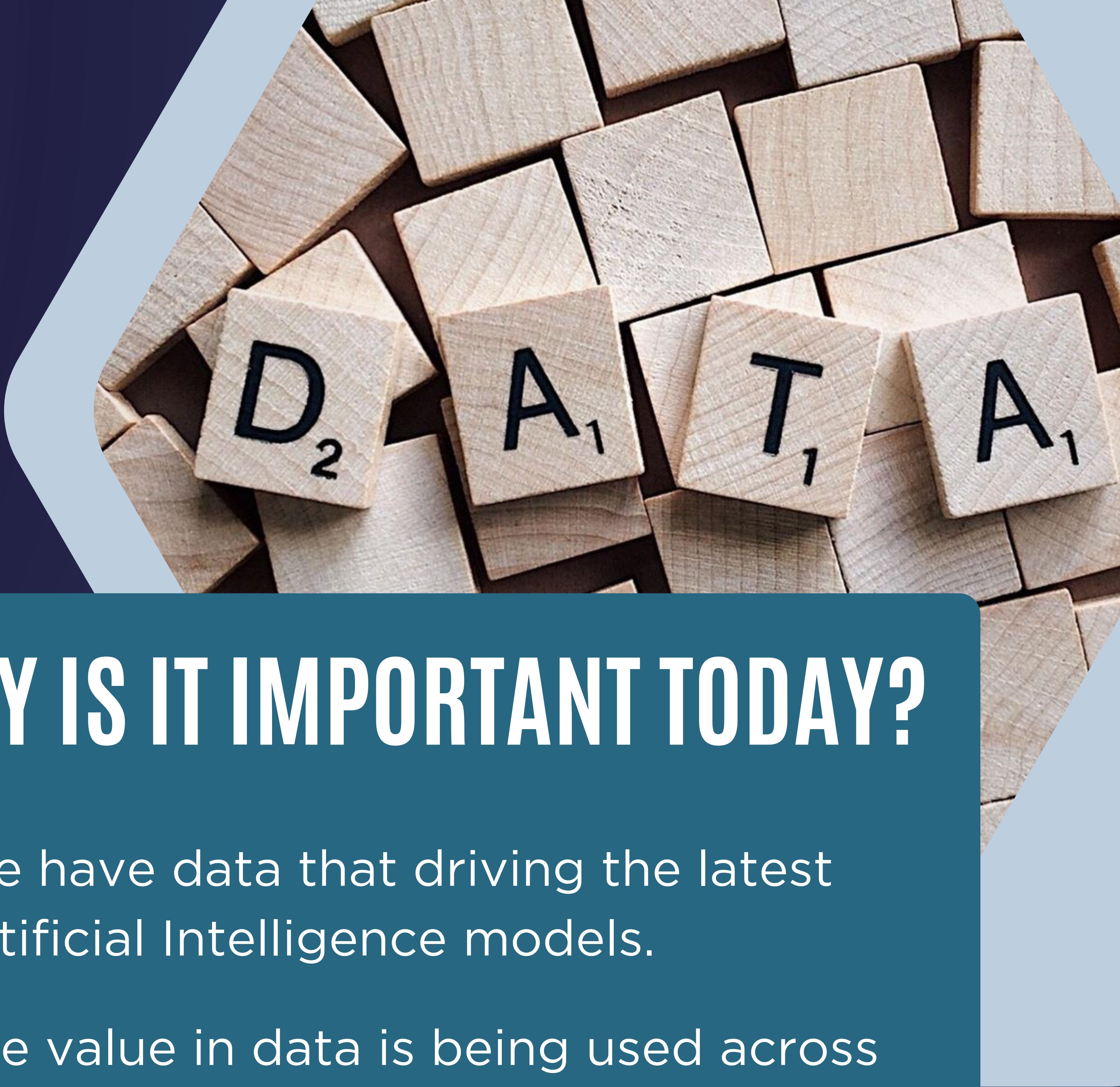
## Data in today's world

# THE IMPORTANCE OF DATA

- In today's world, we have an abundance of data coming in from everywhere.

- Companies want to harness the data and gain relevant insights from the data.



## WHY IS IT IMPORTANT TODAY?

- We have data that driving the latest Artificial Intelligence models.

- The value in data is being used across various industries - Healthcare, finance , marketing.

# CHALLENGES

1

### Variety of data

Data can come to us in a variety of different formats. We could have structured data, semi-structured data or unstructured data.

2

### Rate or Velocity

Not only do we need to consider the data itself but the rate at which the data arrives. How much data is streaming in. How can we manage or handle large inflows of incoming data.

3

### Integrating data

Since we can have data coming in from multiple data streams, how do we integrate the data to get the best out of it.

# CHALLENGES

4

## Infrastructure

What would be the cost for maintaining infrastructure to store the data. What about the costs associated with processing data.

5

## Security

How do we maintain the security of information within data sets. How do we set guard rails to protect the data. How do we manage sensitive information within the data sets.

6

## Governance

We've got data coming in from various sources at different velocities and in different formats. Do we have the right procedures in place when it comes to governance.

# DIFFERENT ROLES

## Data Engineer

The data engineer is responsible for tasks such as maintaining the infrastructure used to get data raw data from incoming sources - clean and make the data relevant

## Data Analyst

The Analyst would analyze the data, generate reports, gain the necessary insights. You can use tools such as Power BI to generate reports and visualizations.

## Data Scientist

This is a professional who would use statistical, mathematical, and computational techniques to extract insights from the structured and unstructured data

## ML Engineers

These are professionals who would build Machine Learning models based on the various available data sets. These Models can be used to solve critical business problems.

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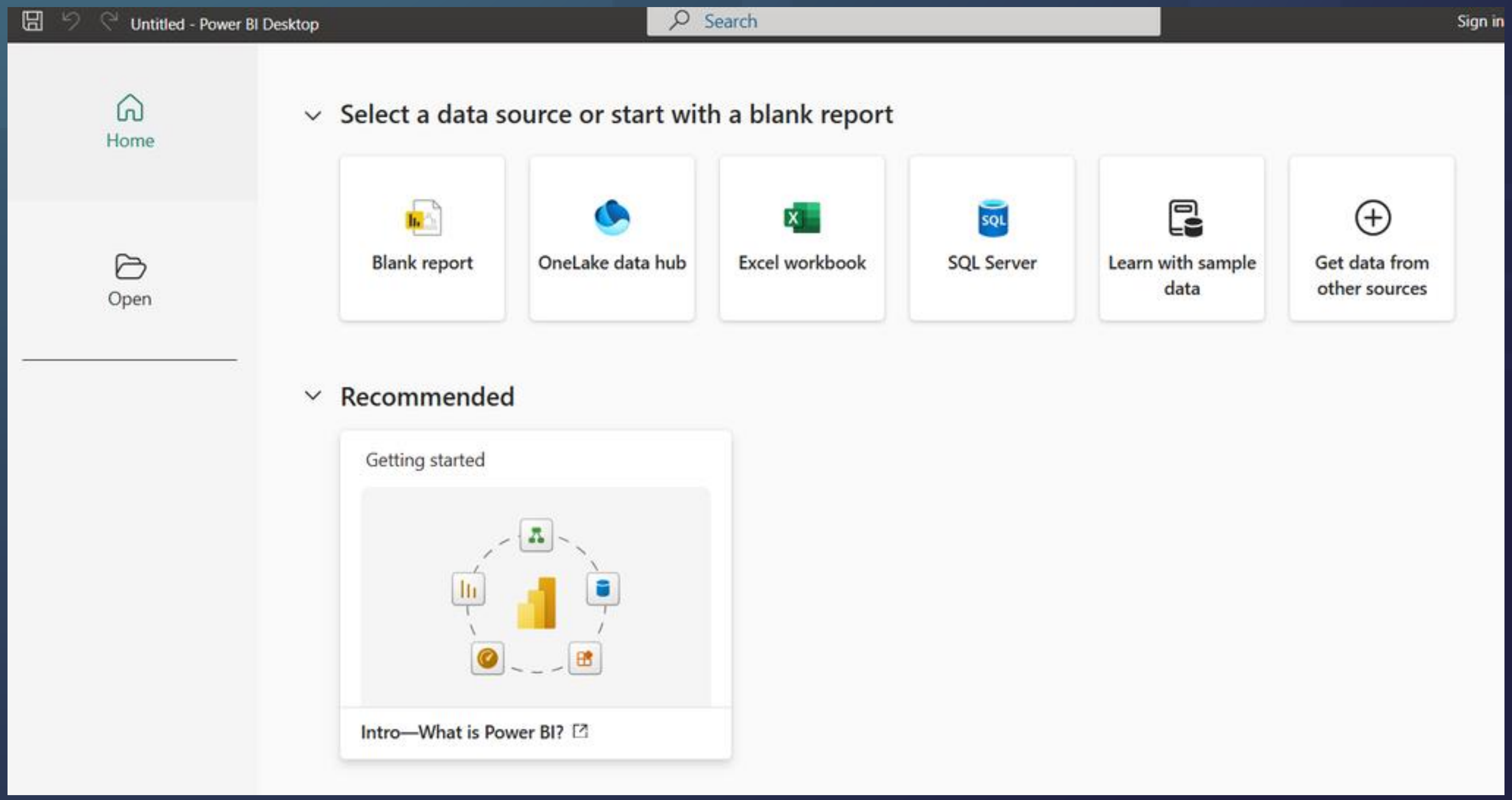
## Microsoft Power BI

# Power BI

- With Power BI you can connect to a variety of data sources.
- You can pull in data from these data sources and start building data models.
- You can then start creating reports and visualizations on top of the data.

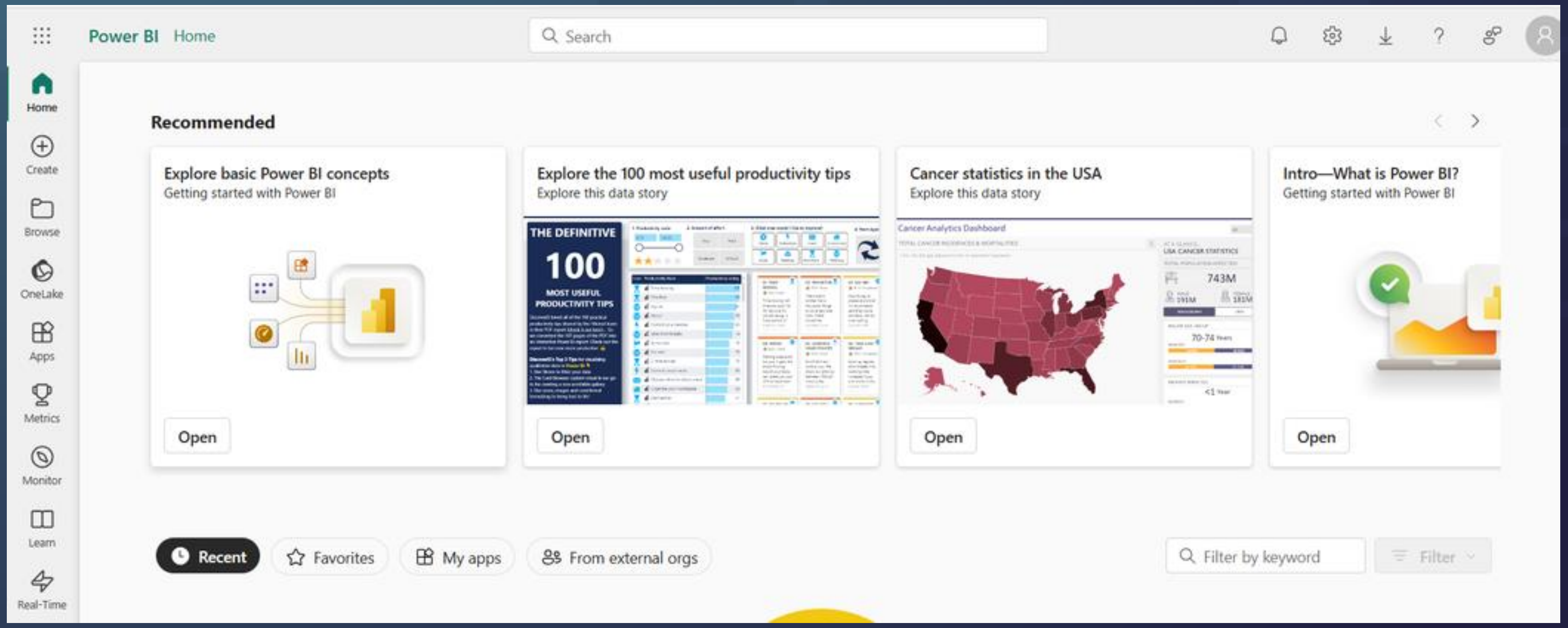
Power BI




# POWER BI DESKTOP

- This is a free application that you can install on your local computer.
- You can use this tool to connect to data sources, fetch data, transform and visualize data.
- You take in data from different sources and build a data model. You can then build reports from the model.

Power BI




# POWER BI SERVICE

- This is an online service available via https://app.powerbi.com
- You can use the browser to consume and interact with reports.
- Once your reports are developed in Power BI desktop, you can publish those reports onto the Power BI service.
- You can also create powerful dashboards via the use of the Power BI service.

# PLANS AND PRICING

Power BI | Microsoft Fabric

### Free account

**Free**

Try for free

Create rich, interactive reports that put visual analytics at your fingertips.

- Included in Microsoft Fabric free account
- No credit card required
- Upgrade to Pro or Premium to share reports

### Power BI Pro

**$10.00**

user/month

Buy now

Access Power BI reports shared with you and publish your own for an even bigger impact.

- Publish and share Power BI reports
- Included in Microsoft 365 E5 and Office 365 E5
- Available to buy now with a credit card[1]

### Power BI Premium Per User

**$20.00**

user/month

Buy now

License data professionals with access to enterprise-scale features.[2]

- Includes all the features available with Power BI Pro
- Access larger model sizes
- More frequent refreshes
- Available to buy now with a credit card[1]

### Power BI Embedded

**Variable**

Contact Sales | Learn more

Create customer-facing reports, dashboards, and analytics in your own applications.

- Brand Power BI reports as your own
- Automate monitoring, management, and deployment
- Reduce developer overhead

Reference - https://www.microsoft.com/en-us/power-platform/products/power-bi/pricing

# PLANS AND PRICING

Power BI | Microsoft Fabric

### Free account

**Free**

Try for free

Give your data teams all the tools they need in a unified, governed, and secure experience.

- No credit card required
- Use trial capacity on any Fabric tool

### Fabric Capacity Reservation

**Variable**

See Fabric pricing

License your organization for access to the end-to-end, unified analytics experience in Fabric.[3]

- Annual purchase, saving 40.5% over pay-as-you-go prices
- Eligible for the Microsoft Azure Consumption Commitment (MACC)
- Per-user licenses required for certain activities[3]

### Fabric Capacity Pay-As-You-Go

**Variable**

See Fabric pricing | Contact Sales

License your organization for access to the end-to-end, unified analytics experience in Fabric.[3]

- Dynamically scale up or down and pause capacity
- Eligible for the Microsoft Azure Consumption Commitment (MACC)
- Per-user licenses required for certain activities[3]

Reference - https://www.microsoft.com/en-us/power-platform/products/power-bi/pricing

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## Implement and Manage semantic models – Power BI Desktop

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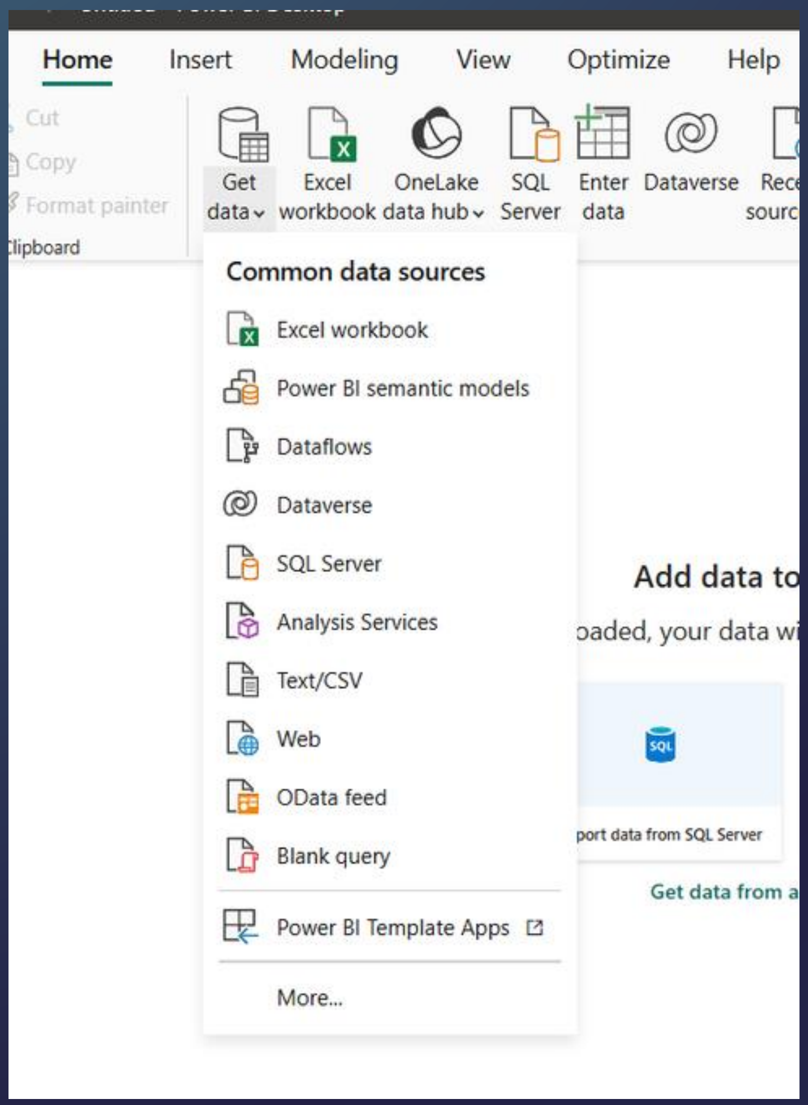
## Connect to data and the Power Query Editor

# CONNECT TO DATA SOURCES

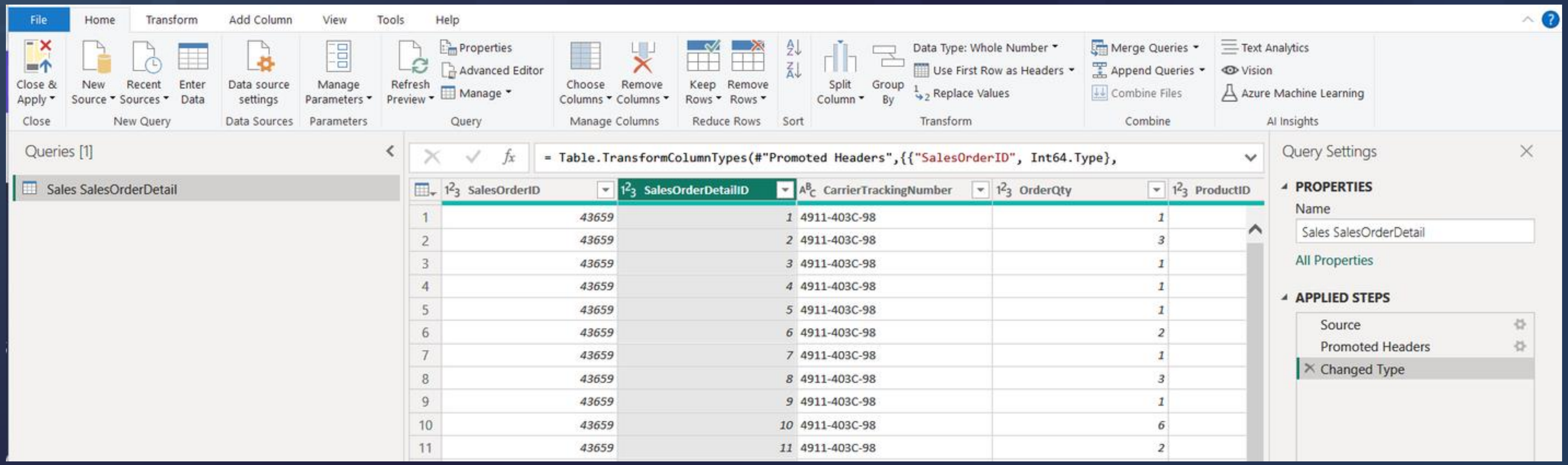
- In Power BI Desktop we can connect to data from different data sources.




1. **File data sources - Text/CSV, JSON**
2. **Databases - SQL Server, Oracle.**
3. **Microsoft Fabric - Dataflows, Warehouses.**
4. **Azure - SQL database, Blob storage.**

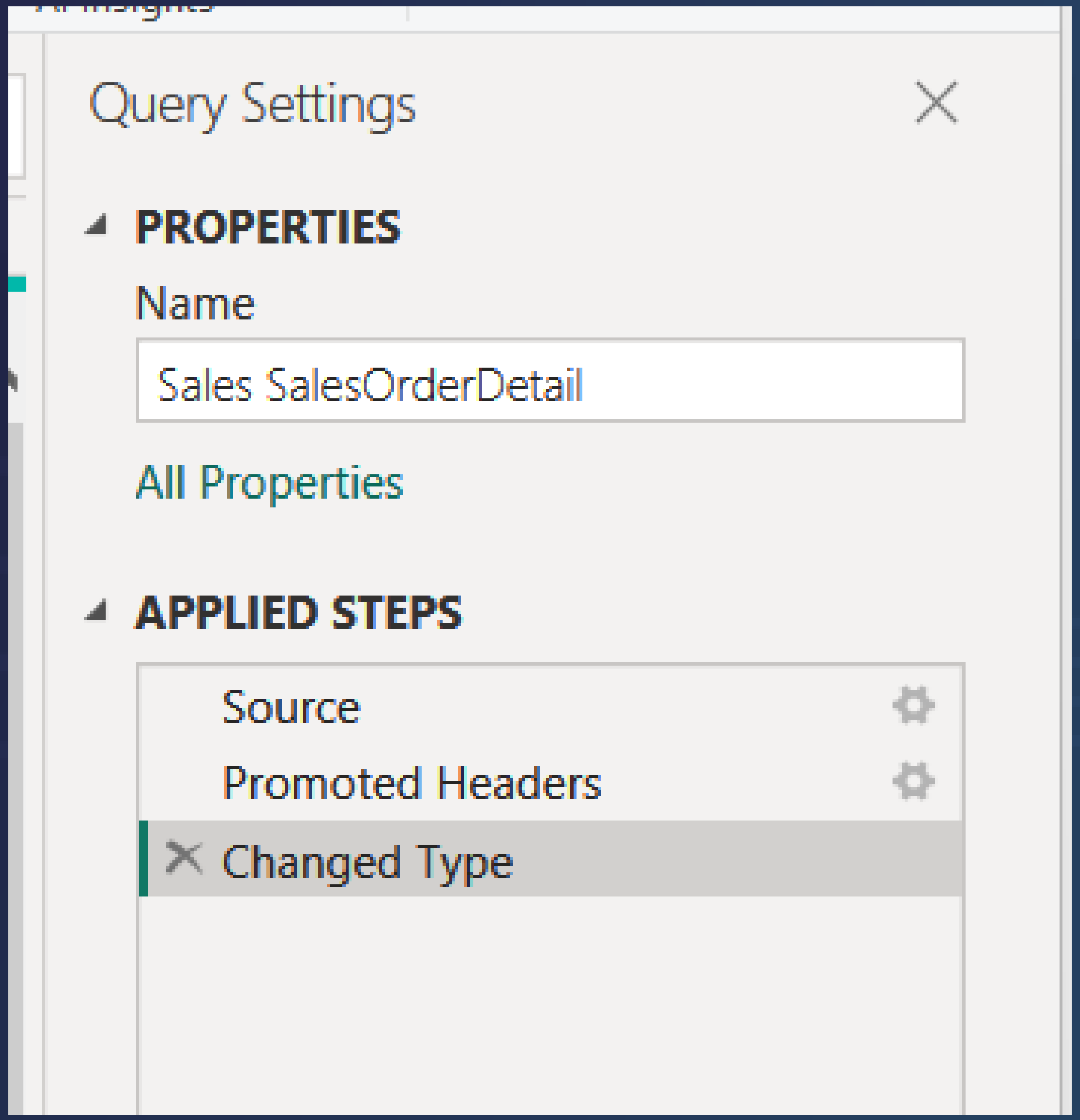
# POWER QUERY EDITOR

- You can use the Power Query Editor to connect to one or more data sources.
- You can then shape and transform your data as required.
- Once your data is in the desired shape, you can load the model in Power BI Desktop.



# POWER QUERY EDITOR

- Once your data is in the Power Query Editor, you start to shape the data as per your needs.
- When you shape the data, you get the instructions which were used to adjust the data.
- This does not change the data in the source. It just changes the view of the data in the Power Query Editor.
- In the Power Query Editor, you might remove columns or rows that are not required. You might rename columns, change column types.



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## DAX

# DAX - DATA ANALYSIS EXPRESSIONS

- This is a formula expression language that can be used in services such as Power BI.
- You can use functions and operators within DAX to perform calculations and queries on your data within your data model.
- DAX can be used to create measures, calculated columns, calculated tables and for row-level security.

# DAX - DATA ANALYSIS EXPRESSIONS

- **Measures** - These are dynamic calculation formulas that provide results based on the context.
- You can add a measure to any table within your model. For this you can make use of a DAX formula. The formula can be used to return a value.
- When it comes to measures, the calculated value does not live within the model. These are calculated at query time.
- You can create a simple measure that could aggregate the values of a single column within your data model.

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## DAX - Aggregation Functions

# DAX - AGGREGATION FUNCTIONS

- COUNT - This is used to count the number of rows in a specified column that contains non-blank values.
- COUNTBLANK - This is used to count the number of blank cells in a column.
- COUNTROWS - This is used to count the number of rows in a table or in a table defined by an expression.
- SUM - This is used to add all of the numbers in a column.
- SUMX - This is used to return the sum of an expression evaluated for each row in a table.
- PRODUCT - This is used to return the product of numbers in a column.
- PRODUCTX - This is used to return the product of an expression evaluated for each row in a table.

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## DAX - FILTER Functions

# DAX - FILTER FUNCTIONS

- CALCULATE - This is used to evaluate an expression in a modified filter context.
- CALCULATETABLE - This is used to evaluate a table expression in a modified filter context.
- FILTER - This is used to return a table that represents a subset of another table or expression.
- RANK - This is used to return the ranking of a row within a given interval.
- RUNNINGSUM - This is used to return the running sum calculated along with the given axis of the visual matrix.
- WINDOW - This is used to return multiple rows which are positioned within a given interval.

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## DAX - INFORMATION Functions

# DAX - INFORMATION FUNCTIONS

- COLUMNSTATISTICS- This returns a table of statistics with regards to every column in the table in the model.
- CONTAINSSTRING - This is used to return a value of TRUE or FALSE depending on whether one string contains another string.
- ISBLANK - This is used to check whether a value is blank or not.
- ISEMPTY- This is used to check if a table is empty.
- ISNUMBER- This is used to check if a value is a number or not.

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## DAX - CALCULATION GROUPS

# DAX - CALCULATION GROUPS

- This helps to effectively reduce the number of measures that you need to create.
- Here you can define a DAX expression that can actually take a selected measure and perform the required calculations.
- You can use calculation groups to reuse complex calculations across your entire data model.

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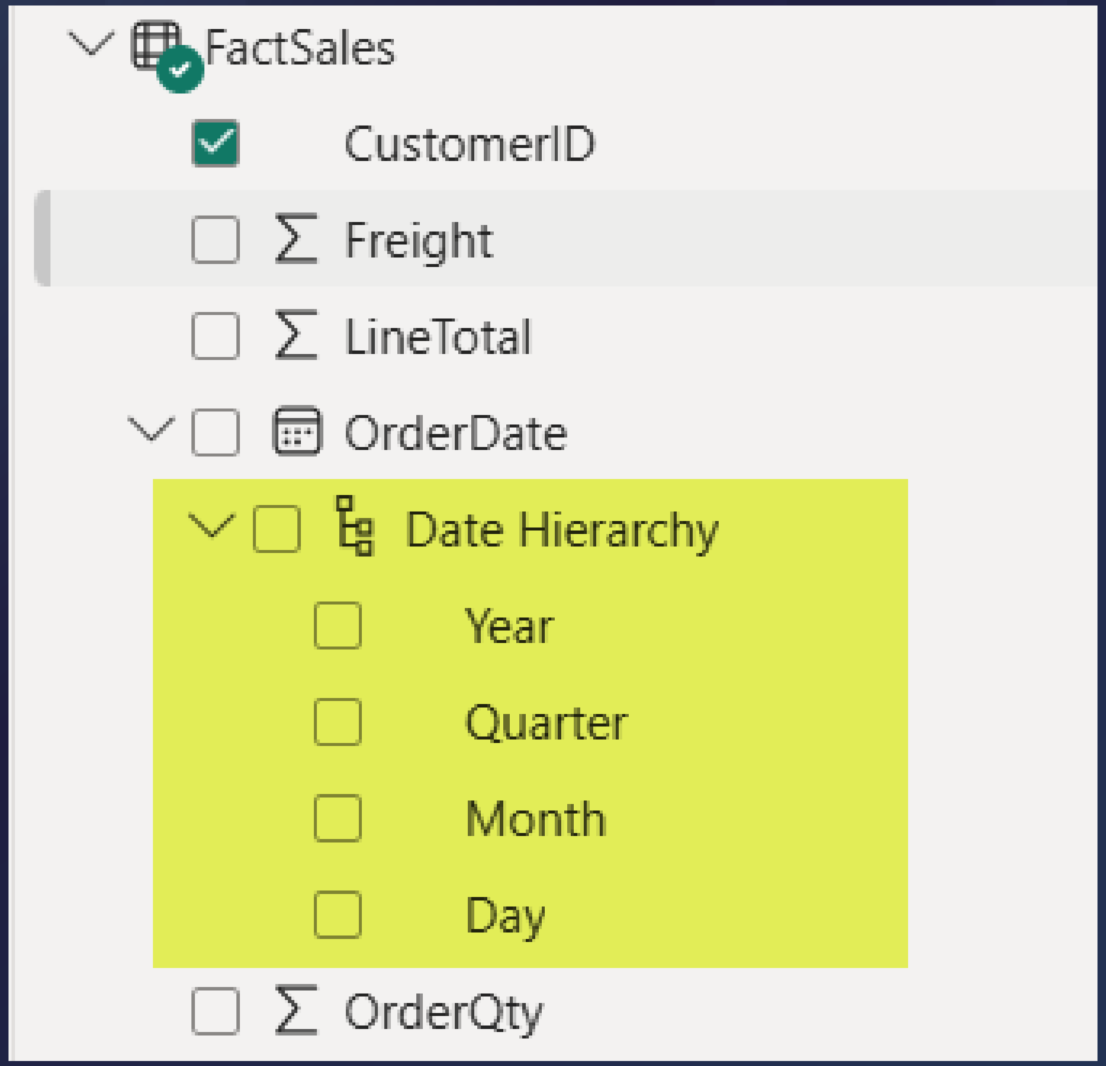
## Power BI Hierarchies

# HIERARCHIES

- This feature allows you to organize related data fields within a hierarchical structure.
- With this feature you can drill down to lower, more detailed levels within your reports.
- When you have a date-based field in your model, Power BI automatically creates a hierarchy for you.



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## Data Profiling tools

# POWER BI - DATA PROFILING

- When using the Power Query Editor, before you can start cleaning and transforming your data, you can make use of data profiling tools to get a better understanding of the values in the column.
- In Power BI, you can get the column distribution, the column profile and the column quality.
- Column distribution - **Distinct** - This gives the overall number of different values in each column.
- Column distribution - **Unique**- This refers to values that only have a single instance of a column.

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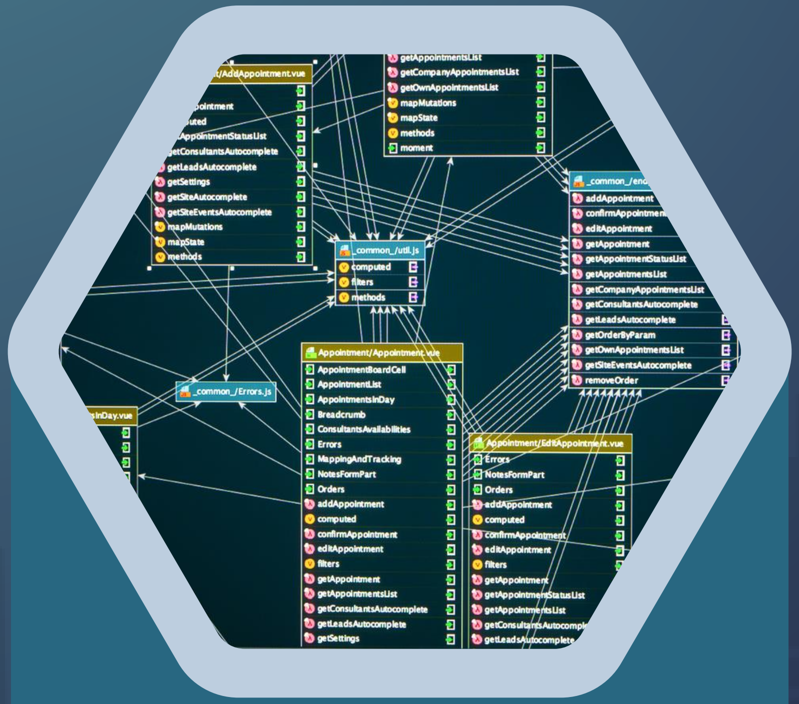
## Storage Mode

# STORAGE MODE FOR DATA MODELS

- **Import mode** - When we used our data sources in Power BI, we used the Import mode to get our data.
- Here we import and store the data within Power BI itself.
- This provides the fastest performance, because your data is already present within Power BI.
- If you need to get the latest data you would need to refresh your model.

# IMPORT MODE - OPTIMIZATION



In your relational data source, define views. Collect the data from the views rather than from the underlying tables.

When taking in data, make sure to only take in data that is required. Remove unnecessary columns and rows.


Make sure the data types are correct. For columns that need calculations, make sure they are based on numerical data types.


You can make use of a feature known as query folding which reduces the work that needs to be done by the underlying Power BI engine.

# STORAGE MODE FOR DATA MODELS

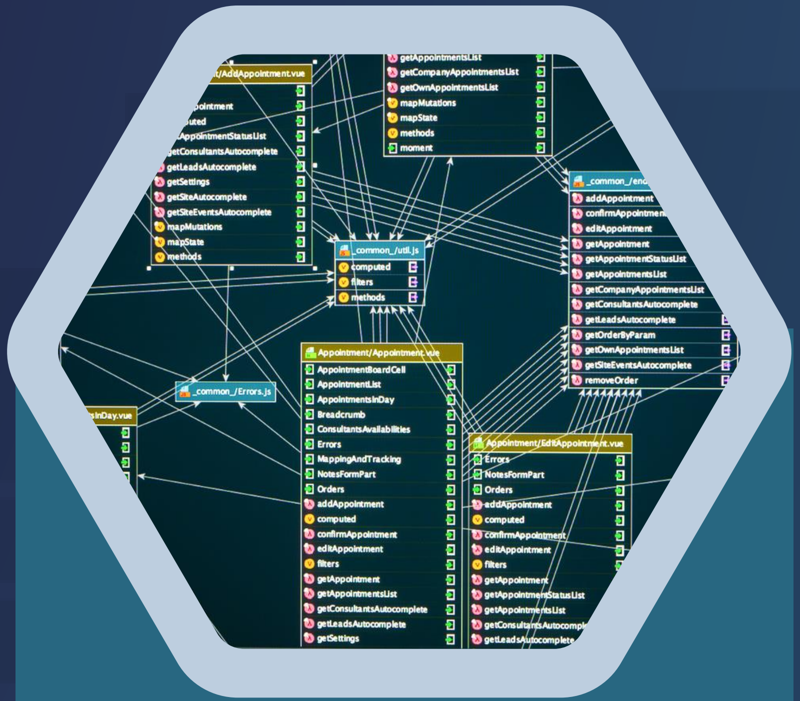
- **DirectQuery Mode** - Here you can query data right from the data source.
- Here the data is not stored within the Power BI service.
- If you have large data sets, this probably would be the ideal mode. Otherwise it would be difficult to manage the data in Power BI via the Import mode.
- You would always get access to the latest data from your source.
- But the performance would be slower when compared with the Import mode.

# DIRECT QUERY - OPTIMIZATION

Try to not add complex calculations and try to use simple DAX expressions.


You can set the Assume referential integrity - This makes use of INNER JOIN rather than OUTER JOIN - Can improve query efficiency.


Add indexes to tables if fetching data from a relational data store. This can help to efficiently retrieve data.



Make sure your dimension tables contain unique values for the key. And that the fact table contains valid dimension key values.

# STORAGE MODE FOR DATA MODELS

- **Direct Lake Mode** - Here you can query data directly from a Microsoft Fabric lakehouse or a warehouse.
- Here again your data is not stored in the semantic model.
- This is also ideal when you have large data sets and when you have a lakehouse or warehouse already setup in Microsoft Fabric.

# STORAGE MODE FOR DATA MODELS

- **Composite Mode** - Here you can mix the modes of Import and DirectQuery.
- This also has support for many-to-many relationships.
- Here you get the benefit of both worlds - You can have some data sources based on the Import mode. And then use other sources for the DirectQuery mode.

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## Performance Analyzer

# POWER BI - PERFORMANCE ANALYZER

- You can use this feature to display the performance of the different elements within your reports when it comes to your visuals and DAX formulas.
- The performance analyzer can measure the processing time to update let's say visuals on your report pages.
- Sometimes users can complain if it takes a long time for reports to refresh. Using the performance analyzer, you can try and pinpoint the probable issue.
- To make sure that you get accurate results when it comes to running the performance analyzer tests, we need to clear the Visual Cache and Data engine Cache.
- For this, we need to create a blank page in the Power BI Desktop file. Then close and start Power BI Desktop again.

# POWER BI - PERFORMANCE ANALYZER


DAX Query - Here the time is recorded from sending the query and getting the results.

Visual Display - This gives the time to render the visual.

Evaluated parameters - time spent in evaluating the field parameters.

There are also other aspects measured like processing for other queries or visuals.

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## Performance Considerations

# POWER BI - PERFORMANCE CONSIDERATIONS

- Choose the right relationship between tables. Power BI would detect relationships between tables. But you can always decide on the cardinality of the relationship between tables - One-to-One (1:1), One-to-many (1:*) , Many-to-one (*:1) and Many-to-many (:).
- Make sure that the required data is only loaded into the model. Otherwise you will have a bloated model and get an undesired performance.
- Your fact tables might have too much granularity in data. Maybe sales are being recorded on a daily basis. You could summarize the data month-wise to reduce the model size.
- Decide on the storage mode - Import vs DirectQuery.

# POWER BI - PERFORMANCE CONSIDERATIONS

- When using DirectQuery , when you load data, only the schema is loaded initially. When you start building visuals then the data is taken from the source.
- If you have multiple users who are using various reports , all based on DirectQuery, then it all depends on how performant your source system is. Else you can get performance issues.
- If you have data coming in from different sources, then you need to consider the performance impact from that aspect as well.

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## Prepare Data – Microsoft Fabric – Data Ingestion and Warehouse

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## Microsoft Fabric Terms

# MICROSOFT FABRIC

- **Experience** - This is a collection of capabilities - Data Engineering, Real-Time Intelligence, Power BI.
- **Workspace** - This is a collection of items in a single environment. Users can collaborate over the items in the workspace.
- **Item** - Each experience in Microsoft Fabric can consist of items. For example, in Data Engineering we can create lakehouses and notebooks.
- **Capacity** - This determines the amount of compute resources allocated to you in Microsoft Fabric.

# MICROSOFT FABRIC - DATA ENGINEERING

- **Fabric Data Warehouse** - This represents a traditional data warehouse that has full support for T-SQL.
- **Lakehouse** - This is a collection files, folders and tables that is used to represent a database over a data lake. It has full support for ACID transactions.
- **Notebook** - This can used to author code and run within a Spark-based environment.

# MICROSOFT FABRIC - DATA FACTORY

- **Connector** - This allows you to connect to a variety of data stores as both the source and destination.
- **Data pipeline** - This is used to orchestrating data movement and transformation.
- **Dataflow Gen2** - This is a low-code interface that allows you to ingest data from different data sources and transform your data.

# MICROSOFT FABRIC - REAL-TIME

- **Eventhouse** - This allows you to manage and analyze large volumes of data in real time.
- **Eventstream** - This provides a centralized place to capture, transform and then route real-time events to different destinations.
- **KQL Database**- You can host a data store that allows you to execute KQL queries.

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## Microsoft Fabric Licensing

# MICROSOFT FABRIC LICENSING

- In order to make use of the different workloads in Microsoft Fabric, we need to have Fabric capacity. This provides the underlying compute to run the workloads.
- A capacity will provide a pool of compute and storage that can used for the various services in Microsoft Fabric.
- You can then assign the capacity to a workspace. The workspace in the end is just an environment where users can collaborate on the different Fabric-based items.

# MICROSOFT FABRIC LICENSING

- For the Microsoft Fabric capacity we have the Pay-as-you-go pricing with the option to reserve capacity as well.

| SKU | Capacity unit (CU) | Pay-as-you-go | Reservation |
|---|---|---|---|
| F2 | 2 | $262.80/month | $156.334/month<br>~41% savings |
| F4 | 4 | $525.60/month | $312.667/month<br>~41% savings |
| F8 | 8 | $1,051.20/month | $625.334/month<br>~41% savings |

Reference - https://azure.microsoft.com/en-us/pricing/details/microsoft-fabric/

# MICROSOFT FABRIC LICENSING

- There is a separate pricing for OneLake Storage. Here remember you are getting a single copy of your data that you can use across all of your Fabric workloads.

| Storage | Price |
|---|---|
| OneLake storage/month** | $0.023 per GB |
| OneLake BCDR storage/month | $0.0414 per GB |
| OneLake cache/month* | $0.246 per GB |

Reference - https://azure.microsoft.com/en-us/pricing/details/microsoft-fabric/

# MICROSOFT FABRIC LICENSING

- You can make use of a Free trial for Microsoft Fabric capacity. This lasts for 60 days.
- You get access to almost all of the workloads and features within Microsoft Fabric.
- With OneLake storage you get access to up to 1 TB of data.
- With the trial capacity we get 64 capacity units.
- You also get a Power BI Individual Trial license. This is equivalent to a Power BI Premium Per user license.

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## Microsoft Fabric - Cloning tables

# MICROSOFT FABRIC - CLONING TABLES

- In the data warehouse, you can clone existing tables.
- Here only the metadata is copied onto a new table.
- The data is not copied. The new cloned table will just reference the same data file in OneLake.
- This can help to reduce storage costs if you want to have multiple cloned tables in place.
- The cloned table can be made available in another schema.

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## Microsoft Fabric - Data pipeline

# MICROSOFT FABRIC - DATA PIPELINE

- The data pipeline allows you to extract, transform and load data from source to a destination.
- This provides a no code approach for performing data ingestion.
- Here the aspects are similar to another service known as Azure Data Factory.
- Azure Data Factory provides a full-fledged cloud managed service for data orchestration.

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## Prepare Data – Microsoft Fabric – Lakehouse

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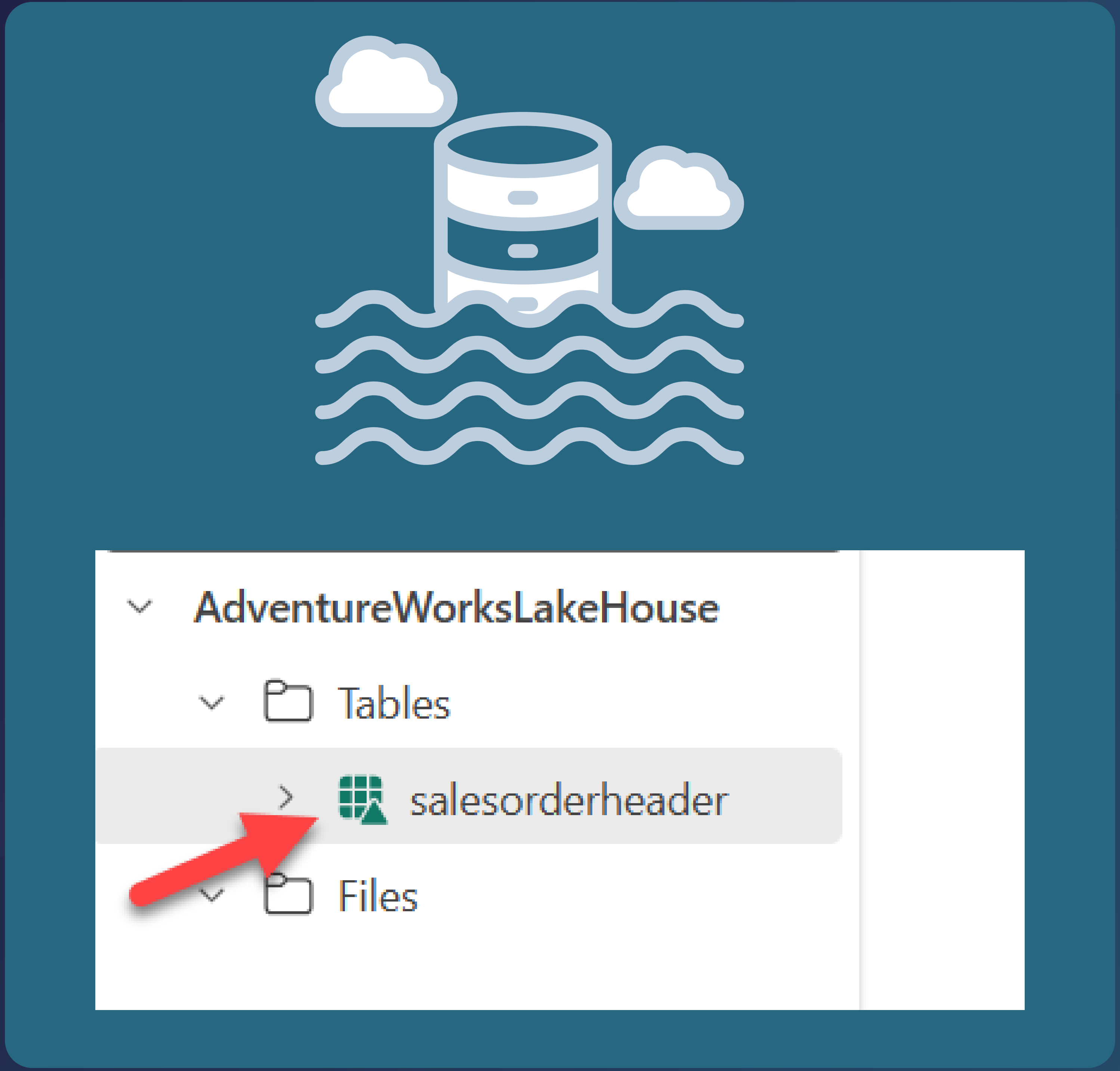
## Delta Lake

# DELTA LAKE

- This is an open-source storage layer.
- Here you have data in a set of files. But on top on that you have the added benefit of database-like capabilities.
- The data files are stored in Delta format. The underlying format is parquet.
- A _delta_Log folder contains the transaction details and this is in JSON format.



# DELTA LAKE

**Relational support**

You can use operations - select, insert, update and delete rows over your data.

**ACID Transactions support**

Atomicity, Consistency, Isolation and Durability.

**Time Travel**

You can track multiple versions of each row within the table.

# DELTA LAKE

- Parquet-based file format
- This is a file format that is stored using the Apache Parquet format. This is a columnar storage format.
- This is a popular format for files across various Big data systems.
- The reading of data from columns is much faster.
- The data type for each column is also stored in a separate layer.
- The files can be compressed using various compression algorithms; hence you can save on storage space.

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## Maintain a data analytics solution

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## Power BI Assets

# POWER BI ASSETS

- When it comes to Power BI Assets, we can include items such as your semantic models, dataflows, reports and dashboards.
- If you have multiple users across your organization who need to create reports based on a semantic model you can ensure that the model is stored in a centralized location.
- You can also build report templates. If there is a particular layout that needs to be in every report, or some standard queries, you can create a standard Power BI template.
- Template files are stored in the format of .pbit.

# POWER BI ASSETS

- Based on the information of the reports in Power BI, a template can contain the following
- Report pages and visual elements.
- The data model definition which includes the schema, relationship, measures etc.
- Query definitions and other query elements.
- The report template will only contain the report definitions and not the report data such. Hence, they will be smaller than the normal Power BI desktop files.
- Once you have the report template in place, you can import the template in Power BI Desktop.

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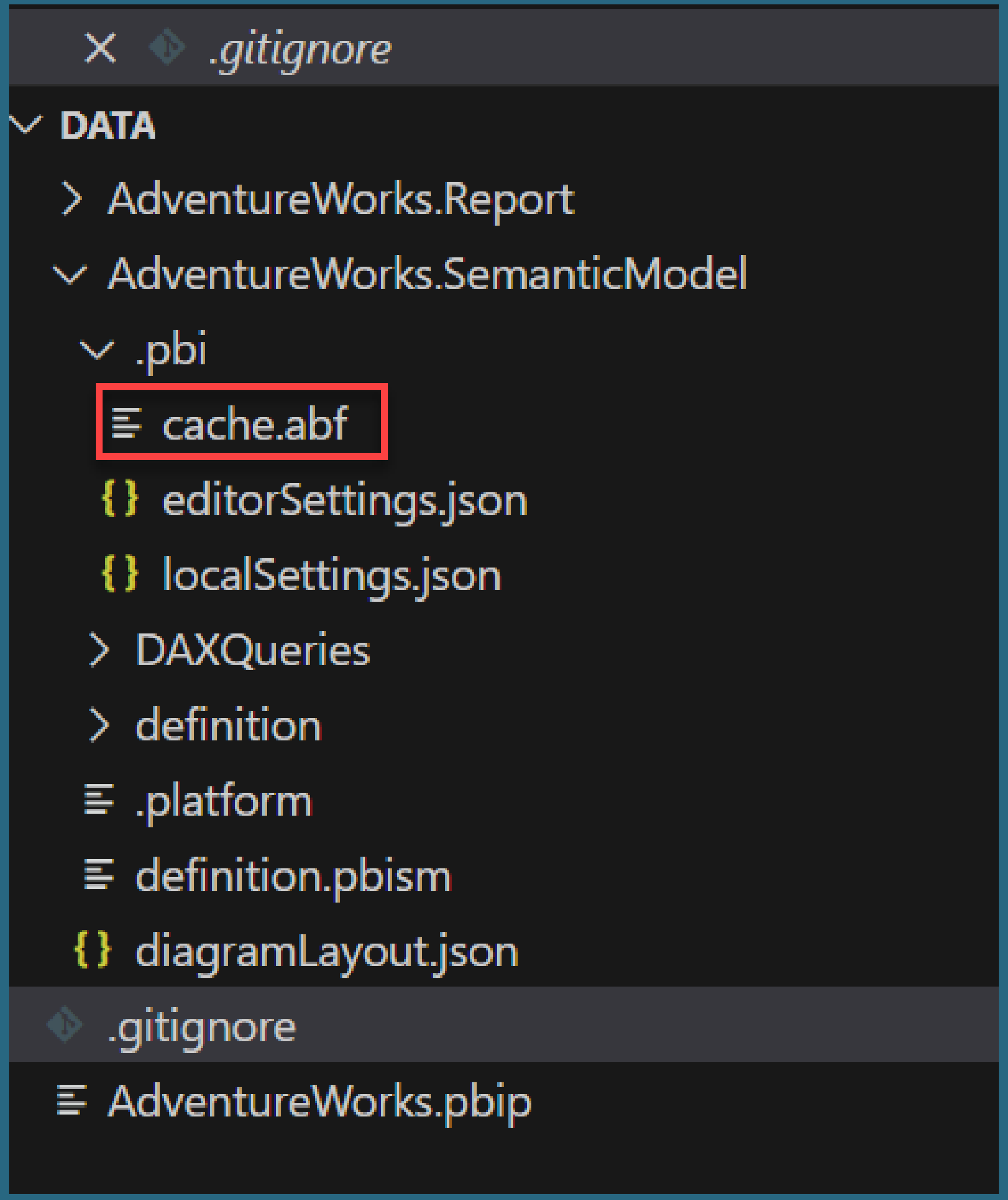
## Power BI Project files

# POWER BI PROJECT FILES

- When it comes to Power BI Project files, they are saved with an extension of .pbip. When it comes to Power BI Desktop files, they are saved as .pbix files and are just binary files.
- With Power BI Project files, the semantic model and reports are stored in individual plain text files. These allow the files to be used with code editors such as Visual Studio Code.
- You can also make use of version control systems to manage multiple versions of the Power BI Project files.

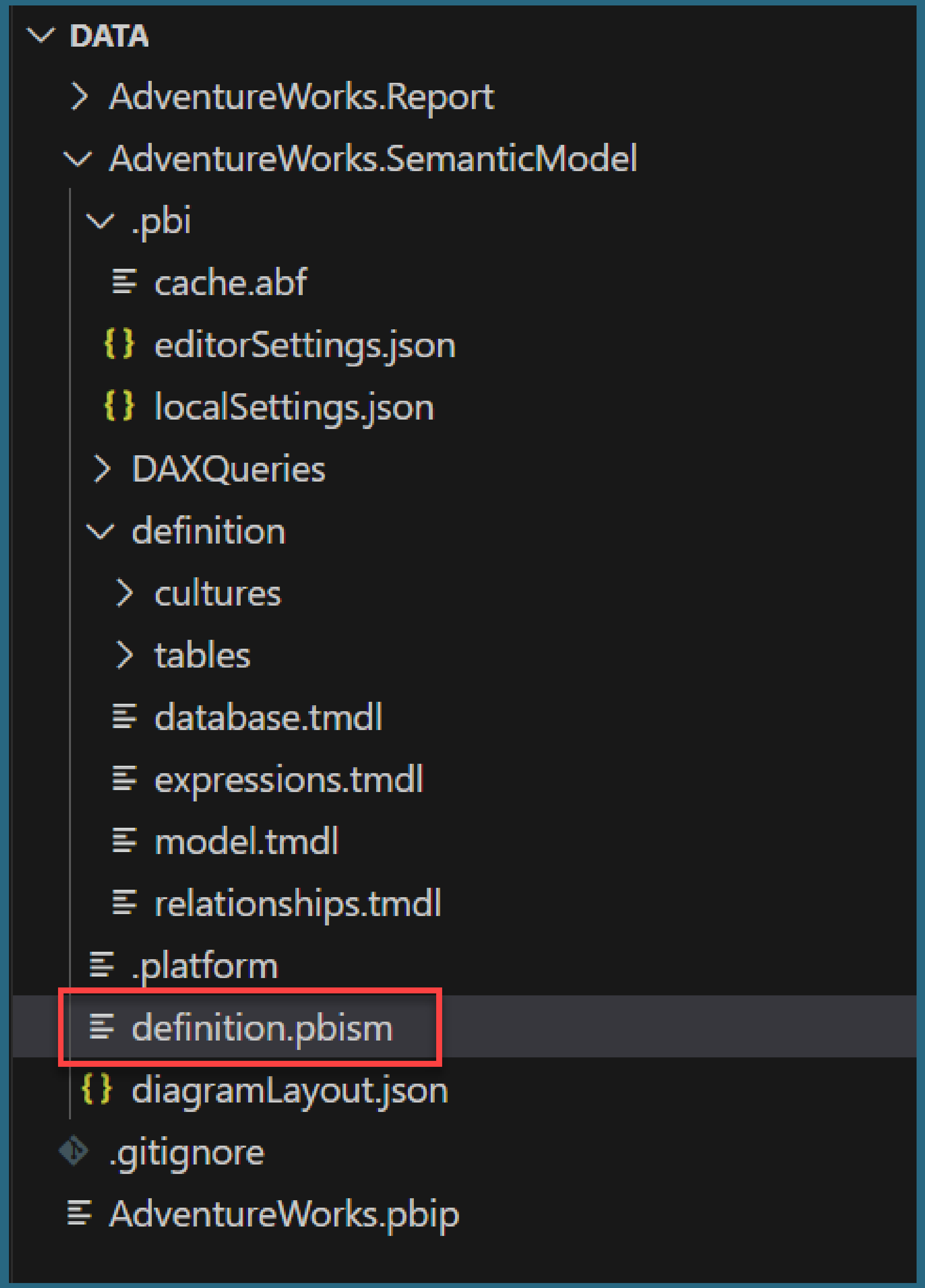
# POWER BI PROJECT FILES

- .pbi\cache.abf - This is a file containing the local cached copy of the model and the data based on when it was last edited.
- This file should not be checked into source control. By default the git will ignore this file.



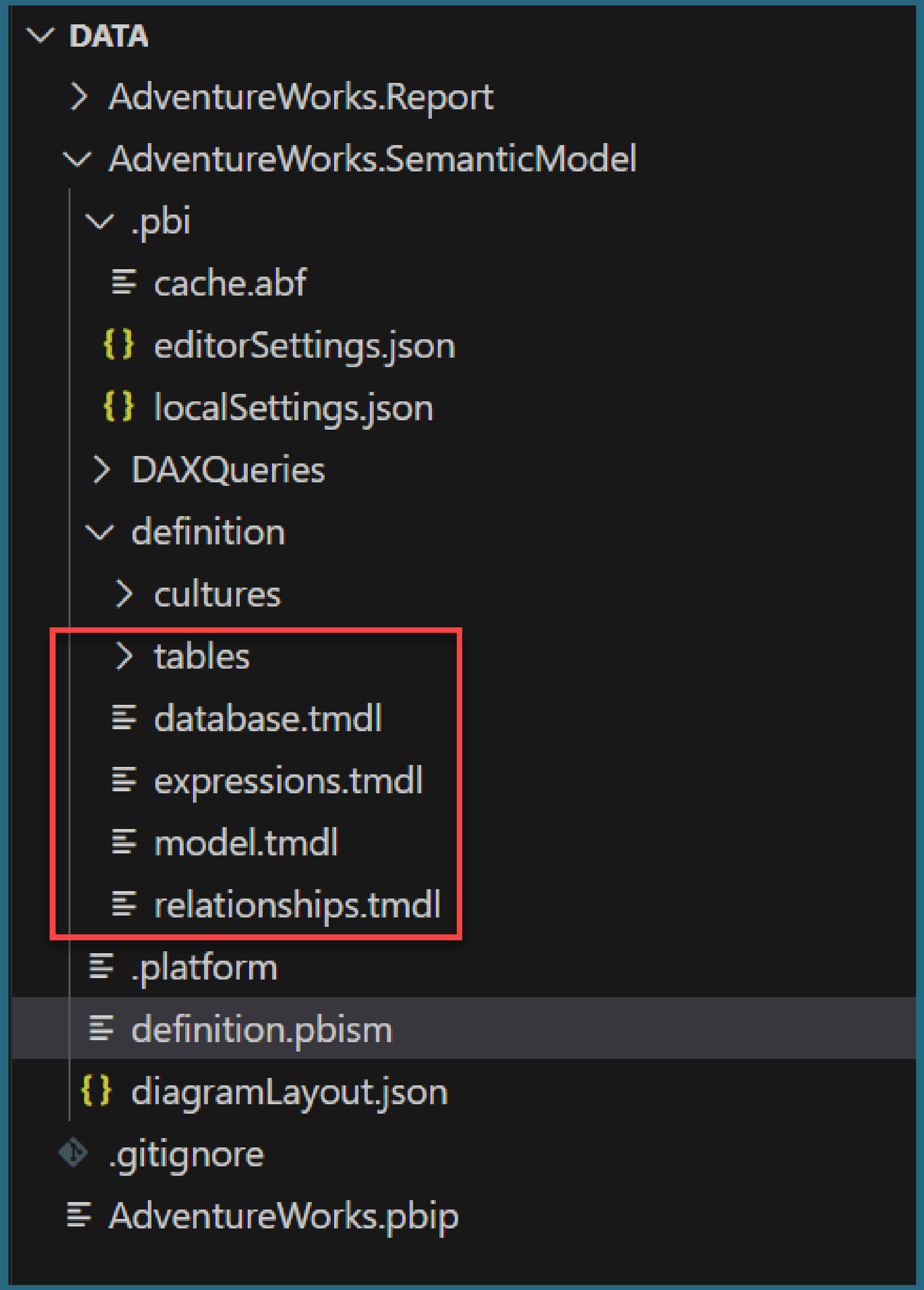
# POWER BI PROJECT FILES

- definition.pbism - This is a file containing the overall definition of the semantic model and the core settings.
- If the version is 4.0, then the semantic model definition can be stored as TMSL or TMDL.



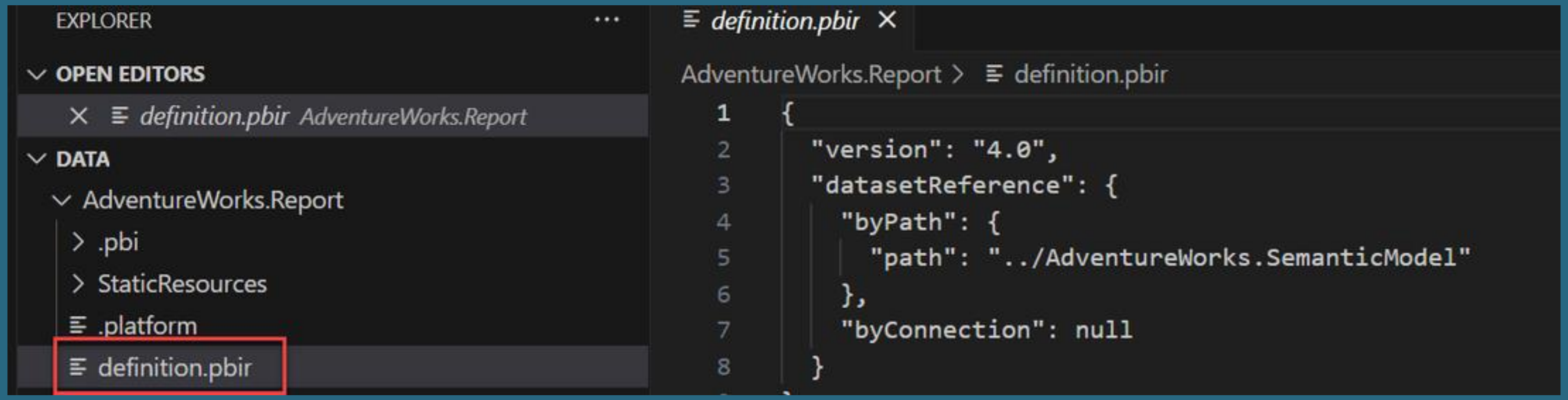
# POWER BI PROJECT FILES

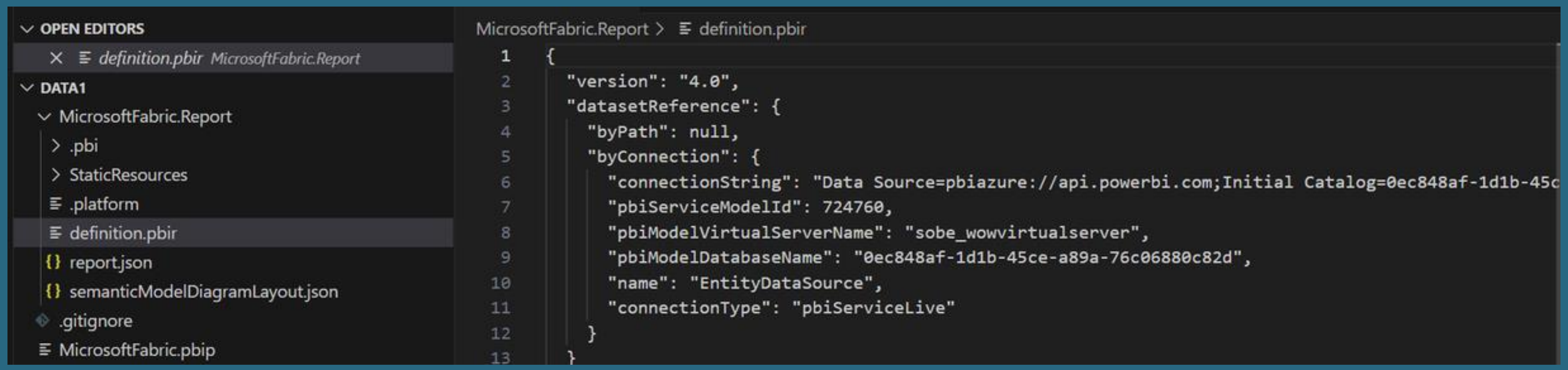
- The semantic models can now be saved in Tabular Model Definition Language (TMDL). This is designed to be easily readable and editable as per its older counterpart - (TMSL) - Tabular Model Scripting Language.



# POWER BI PROJECT FILES

- Then we have the files in the Project report folder.
- definition.pbir - This contains the overall definition of a report and the core settings.
- Currently this is using the most current version of this file.
- If there is a byPath value mentioned then Power BI will open the semantic model in edit mode.
- If there is a reference to a remote semantic model in the Power BI service, you would see a value for the byConnection key.



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## Data warehouse security

# DATA WAREHOUSE SECURITY

- There are different Microsoft Fabric roles available for the data warehouse
- Read permission - Here the user will only be able to connect to the SQL Analytics endpoint. The user will not be able fire any queries against the tables in the warehouse.
- ReadData permissions - This is equivalent to the db_datareader role in SQL Server that allows users to read data from the tables and the views in the warehouse.
- ReadAll permissions - This gives permissions to the user to read all of the parquet files in OneLake that is consumed by using Spark.

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## Sensitivity Labels

# SENSITIVITY LABELS

- This allows users to classify sensitive and critical content.
- Sensitivity labels can be applied in Microsoft Fabric and Power BI.
- In Power BI, they can be applied at the desktop level and also in the Power BI service.
- You can apply sensitivity labels to semantic models, reports, dashboards and data flows.
- The sensitivity labels are applied from Microsoft Purview Information Protection.

# SENSITIVITY LABELS

- In order to make use of sensitivity labels, we first need to enable it at the tenant level.
- To apply labels to Power BI Fabric items, the user needs to have Power BI Pro or Premium Per user license.
- They also need an Azure Information Protection Premium P1 or Premium P2 license.

# DP-600

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## Endorse items

# ENDORSE ITEMS

- In Microsoft Fabric, you can endorse your high-quality items. This increases their visibility. There are three ways to achieve this.
- **Promotion** - Here you want to highlight items that are valuable and would be worthwhile for other users in your organization.
- **Certification** - Here this means that the item has met the quality standards of the organization.
- **Master data** - Here the item is regarded as the single-source-of-truth-data.